

FAULT CODE 118 - Fuel Pump Pressure Sensor Circuit

Warnings and Cautions

⚠ CAUTION ⚠

To reduce the possibility of damaging a new engine control module (ECM), all other active fault codes must be investigated prior to replacing the ECM.

⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3824774 - Packard™ breakout cable Part Number 3164752 - DIN breakout cable.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check for multiple fault codes.	
	STEP 1A. Read the fault codes.	Fault Codes 116 and 451 are not active?
STEP 2.	Check the fuel pump pressure sensor.	
	STEP 2A. Inspect the fuel pump pressure sensor and engine harness connector pins.	Dirty or damaged pins?
	STEP 2B. Read the fault codes.	Fault Code 118 active?
	STEP 2C. Check the fuel pump pressure supply voltage.	4.75 to 5.25-VDC?

STEPS		SPECIFICATIONS
	STEP 2C-1. Check the fuel pump pressure supply voltage.	4.75 to 5.25-VDC?
	STEP 2D. Check the fuel pump pressure signal voltage.	0.42 to 0.58-VDC engine stopped?
	STEP 2D-1. Check for a short circuit from pin to pin.	More than 100k ohms"
	STEP 2D-2. Check for continuity in the engine harness.	Less than 10 ohms?
	STEP 2D-3. Check the ECM response.	Fault Code 118 inactive; Fault Code 119 active?
	STEP 2D-4. Check the resistance from signal pin to supply pin in the ECM port.	More than 35k ohms?
STEP 3.	Check the engine harness.	
	STEP 3A. Inspect the engine harness and ECM connector pins.	Dirty or damaged pins?
	STEP 3B. Read the fault codes.	Fault Code 118 active?
	STEP 3C. Check for an open circuit.	Less than 10 ohms?
	STEP 3D. Check for a short circuit from pin to pin.	More than 100k ohms?
STEP 4.	Check for an ECM response.	
	STEP 4A. Check for the appropriate ECM response.	Fault Code 118 inactive; Fault Code 119 active?
STEP 5.	Clear the fault codes.	
	STEP 5A. Disable the fault code.	Fault Code 118 inactive?
	STEP 5B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check for multiple fault codes.

STEP 1A. Read the fault codes.

Conditions :

- Connect INSITE™ electronic service tool
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
Read the fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Fault Codes 116 and 451 are not active?	2A
		Multiple fault code tree

STEP 2. Check the fuel pump pressure sensor.

STEP 2A. Inspect the fuel pump pressure sensor and engine harness connector pins.

Conditions :

- Turn keyswitch OFF
- Disconnect the engine harness from the fuel pump pressure sensor.

Action	Specification/Repair	Next Step
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Inspect fuel pump pressure sensor and engine harness connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html) •	Dirty or damaged pins?	2B
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A damaged connection has been detected in the ECM engine connector or engine harness connector.

Repair damaged pins.

Repair or replace the engine harness, or replace the fuel pump pressure sensor, whichever has the damaged pins.

- Flush the dirt, debris, or moisture from the connector pins, use electronic contact cleaner, Part Number 3824510.
- Install the appropriate connector seal if it is damaged or missing.
- Repair the engine harness. Refer to Procedure 019-201 in Section 19.
(/qs3/pubsys2/xml/en/procedures/99/99-019-201.html)
- Refer to Procedure 019-202 in Section 19.
(/qs3/pubsys2/xml/en/procedures/99/99-019-202.html)
- Replace the engine harness. Refer to Procedure 019-043 in Section 19.
(/qs3/pubsys2/xml/en/procedures/19/19-019-043.html)
- Replace the timing pressure sensor. Refer to Procedure 019-191 in Section 19.

	(/qs3/pubsys2/xml/en/procedures/19/19-019-191.html) Replace the fuel pump pressure sensor. Refer to Procedure 019-118 in Section 19. (/qs3/pubsys2/xml/en/procedures/00/00-019-118.html)	
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STEP 2B. Read the fault codes.

Conditions :

- Connect all components
- Connect INSITE™ electronic service tool
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
Read the fault codes. Use INSITE™ electronic service tool to read the fault codes.	Fault Code 118 active?	2C
	Repair complete.	5B

STEP 2C. Check the fuel pump pressure supply voltage.

Conditions :

- Disconnect the fuel pump pressure sensor from the engine harness
- Install the appropriate fuel pump pressure sensor breakout cable between the sensor and the engine harness connector
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
Check the fuel pump pressure supply voltage. Measure the supply voltage by installing the breakout cable's supply (pin A) and return (pin B) connectors into the multimeter.	4.75 to 5.25-VDC?	2D
		2C-1

STEP 2C-1. Check the fuel pump pressure supply voltage.

Conditions :

- Disconnect the breakout cable from the sensor. (Leave breakout cable connected to the engine harness.)
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
Check the fuel pump pressure supply voltage. • Measure the supply voltage by installing the breakout cable's supply (pin A) and return (pin B) connectors into the multimeter.	4.75 to 5.25-VDC? Replace the fuel pump pressure sensor. Refer to Procedure 019-118 in Section 19. (/qs3/pubsys2/xml/en/procedures/00/00-019-118.html)	5A
		3A

STEP 2D. Check the fuel pump pressure signal voltage.

Conditions :

- Disconnect the fuel pump pressure sensor from the engine harness
- Install the appropriate fuel pump pressure sensor breakout cable between the sensor and the engine harness connector
- Turn keyswitch ON
- Engine **not** running.

Action	Specification/Repair	Next Step
Check the fuel pump pressure signal voltage. • Measure the signal voltage by installing the breakout cable's signal (pin C) and return (pin B) connectors into the multimeter.	0.42 to 0.58-VDC?	3A
	Does not meet specification.	2D-1

STEP 2D-1. Check for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF
- Disconnect the engine harness from the ECM
- Disconnect fuel pump pressure sensor from the engine harness
- Disconnect rail pressure sensor from the engine harness
- Disconnect the timing pressure sensor from the engine harness
- Disconnect the barometric pressure sensor from the engine harness.

Action	Specification/Repair	Next Step
Check for a short circuit from pin to pin. • Measure the resistance from pin 5 of the engine harness ECM connector to all other pins in the connector. • Measure the resistance from pin 32 of the engine harness ECM connector to all other pins in the connector. • Measure the resistance from pin 18 of the engine harness ECM connector to all other pins in the connector.	More than 100k ohms?	2D-2
	Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-204 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html) • Replace the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-043.html)	5A

STEP 2D-2. Check for continuity in the engine harness.**Conditions :**

- Turn keyswitch OFF
- Disconnect the engine harness from the ECM
- Disconnect the fuel pump pressure sensor from the engine harness.

Action	Specification/Repair	Next Step
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Check for continuity in the engine harness. • Measure the resistance from pin 5 of the engine harness ECM connector to pin A of the fuel pump pressure sensor connector. • Measure the resistance from pin 32 of the engine harness ECM connector to pin C of the fuel pump pressure sensor connector. • Measure the resistance from pin 18 of the engine harness ECM connector to pin B of the fuel pump pressure sensor connector.	Less than 10 ohms?	2D-3
	Repair or replace the engine harness • Repair the engine harness. Refer to Procedure 019-201 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-201.html) • Refer to Procedure 019-202 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html) • Refer to Procedure 019-204 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html) • Replace the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-043.html)	5A

STEP 2D-3. Check the ECM response.

Conditions :

- Disconnect the engine harness from the ECM
- Connect INSITE™ electronic service tool
- Turn keyswitch ON.

Action	Specification/Repair	Next Step

Check for the appropriate ECM response. Use INSITE™ electronic service tool to read the fault codes.	Fault Code 118 inactive; Fault Code 119 active?	2D-4
	Fault Code 118 active. Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-031.html)	5A

STEP 2D-4. Check the resistance from signal pin to supply pin in the ECM port.

Conditions :

- Turn keyswitch OFF
- Disconnect the engine harness connector from the ECM.

Action	Specification/Repair	Next Step
Check the resistance from signal pin to supply pin in the ECM port. Measure the resistance from pin 32 (signal) to pin 5 (supply) in the ECM engine harness connector port.	More than 35 ohms? Replace the fuel pump pressure sensor. Refer to Procedure 019-118 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-118.html)	5A
	Less than 35k ohms? Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-031.html)	5A

STEP 3. Check the engine harness.

STEP 3A. Inspect the engine harness and ECM connector pins.

Conditions :

- Turn keyswitch OFF
- Disconnect the engine harness from the ECM
- Disconnect the engine harness from the fuel pump pressure sensor.

Action**Specification/Repair****Next Step**

Inspect the engine harness and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris in or on the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html) •	Dirty or damaged pins?	3B
	A damaged connection has been detected in the ECM engine connector or engine harness connector. Repair damaged pins. Repair or replace the engine harness, or replace the ECM, whichever has the damaged pins. • Flush the dirt, debris, or moisture from the connector pins, use electronic contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. • Repair the engine harness. Refer to Procedure 019-204 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html) • Replace the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-043.html) • Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-031.html)	5A

STEP 3B. Read the fault codes.

Conditions :

- Connect all components
- Connect INSITE™ electronic service tool
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
Read the fault codes. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 118 active?	3C
	Repair complete.	5B

STEP 3C. Check for an open circuit.**Conditions :**

- Turn keyswitch OFF
- Disconnect the engine harness from the ECM
- Disconnect the engine harness from the fuel pump pressure sensor.

Action	Specification/Repair	Next Step
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	Less than 10 ohms?	3D
	Repair or replace engine harness. • Repair the engine harness. Refer to Procedure 019-201 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-201.html) • Refer to Procedure 019-202 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html) • Refer to Procedure 019-204 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html) • Replace the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-043.html)	5A

Check for an open circuit.

- Measure the resistance from pin 18 of the engine harness connector to pin B of the fuel pump pressure sensor harness connector, engine harness side.

STEP 3D. Check for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF
- Disconnect the engine harness from the ECM
- Disconnect the engine harness from the timing pressure sensor
- Disconnect the engine harness from the rail pressure sensor
- Disconnect the engine harness from the fuel pump pressure sensor
- Disconnect the barometric pressure sensor from the engine harness.

Action	Specification/Repair	Next Step
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Check for short circuit from pin to pin. • Measure the resistance from pin 5 of the engine harness connector to all other pins in the connector. • Measure the resistance from pin 32 of the engine harness connector to all other pins in the connector. • Measure the resistance from pin 18 of the engine harness connector to all other pins in the connector. Look for short circuits from battery or sensor voltage to pin 5, 32, or 18.	More than 100k ohms?	4A
	Repair or replace engine harness • Repair the engine harness. Refer to Procedure 019-204 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html) • Replace the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-043.html)	5A

STEP 4. Check for an ECM response.

STEP 4A. Check for the appropriate ECM response.

Conditions :

- Disconnect the engine harness from the ECM
- Connect INSITE™ electronic service tool
- Turn keyswitch ON.

Action	Specification/Repair	Next Step

Check for the appropriate ECM response. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 118 inactive; Fault Code 119 active?	5A
	• Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/19/19-019-031.html)	5A

STEP 5. Clear the fault codes.

STEP 5A. Disable the fault code.

Conditions :

- Connect all components
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable the fault code. • Start the engine and let it idle for 1 minute. • Use INSITE™ electronic service tool to verify that Fault Code 116 is inactive.	Fault Code 118 inactive?	5B
	Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 5B. Clear the inactive fault codes.

Conditions :

- Connect all components
- Turn keyswitch ON
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared?	Repair complete
	Troubleshoot any remaining active fault codes.	Appropriate troubleshooting charts

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